

Qatar-1b

R-2780

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_140616 · created 2026-09-03T23:56:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456824.8338 +/- 0.00032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.116 +/- 0.055
Transit depth (Rp/Rs)^2	1.34 +/- 1.28 [%]
Orbital Inclination (inc)	89.0 +/- 2.98
Airmass coefficient 1 (a1)	1.54 +/- 0.17
Airmass coefficient 2 (a2)	-0.32 +/- 0.36
Scatter in the residuals of the lightcurve fit is	11.2 %
Best Comparison Star	#2 - [577, 111]
Optimal Method	PSF photometry
Transit Duration (day)	0.077 +/- 0.0075

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.116 ± 0.055 vs 0.1463 (deviation 0.55σ)
Duration fit vs published	0.077 d vs 0.0692 d (deviation 1.04σ)
Mid-time O-C	0.7 min
Depth SNR	2.1
Residual scatter	11.2 %

Light curve

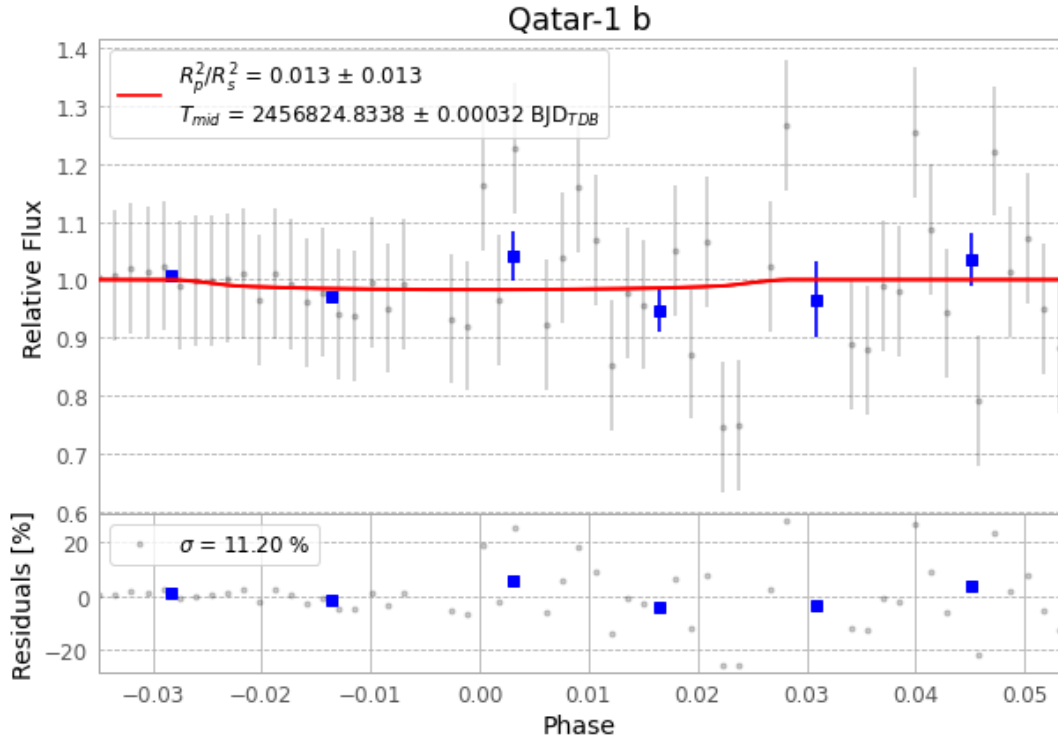


Figure 1 — FinalLightCurve_Qatar-1 b_2014-06-15.png (SHA-256 80e2db250b11578a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [390, 270], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f8d041d809655fbd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

59 FITS frames (39 MB), Qatar-1140616064812.FITS → Qatar-1140616094813.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-140616.log (4b0d50df7cdd...), FinalLightCurve_Qatar-1 b_2014-06-15.png (80e2db250b11...), FinalLightCurve_Qatar-1 b_2014-06-15.csv (e5ec23bccc3a...)

Compute resources

Wall time 186.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 23:56Z.